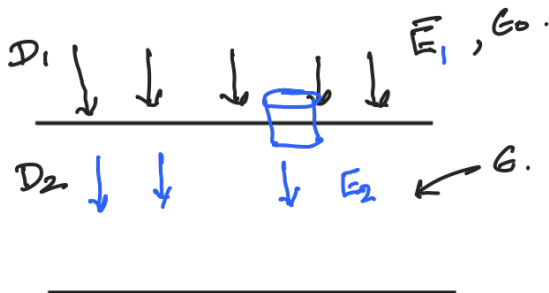


Dielectric-Dielectric Boundary Conditions in Electrostatics.

If we apply Gauss's law
in $\vec{E}$

$$\vec{E}_{ind} = \frac{\sigma_b}{\epsilon_0} = E_1 - E_2$$

$$\Rightarrow E_1 - E_2 = \frac{\sigma_b}{\epsilon_0} ?$$

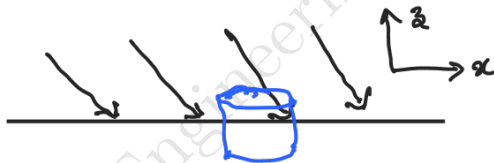
But if we apply Gauss's law in $\vec{D}$

$$\nabla \cdot \vec{D} = \rho_f = 0 \quad (\text{as } \rho_f = 0 \text{ here}).$$

ie, $D_1 = D_2$ (Displacement vector does not see any charge here).

$$\text{ie, } \epsilon_0 E_1 = \epsilon_0 E_2$$

Now if we have $\vec{E}$ "not normal" to the boundary,
By Gauss's law,



$$E_{1z} - E_{2z} = \frac{\sigma_b}{\epsilon_0}$$

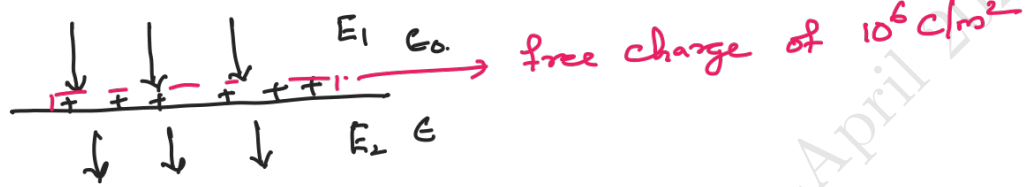
$$\text{Similarly, } D_{1z} - D_{2z} = 0$$

$$\Rightarrow D_{1z} = D_{2z}$$

Boundary
Condition (B.C)
(1)

ie, Normal components of displacement vector is continuous at the dielectric boundary which is free of free charge.

Suppose we have free charge (10^6 C/m^2) at the dielectric boundary; thus

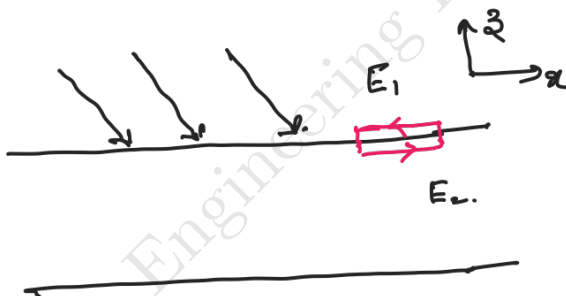


Then the displacement vector is discontinuous by that much of free charge.

$$\text{ie; } D_1 - D_2 = 10^6 \text{ C/m}^2.$$

We know that $\oint \vec{E} \cdot d\vec{l} = 0$

Apply this in the dielectric boundary.



Consider a loop with vertical component $\vec{z}$ negligible.

Apply the condition $\oint \vec{E} \cdot d\vec{l} = 0$

$$\Rightarrow -E_1 \Delta l + E_2 \Delta l = 0$$

$$\Rightarrow E_1 \Delta l = E_2 \Delta l$$

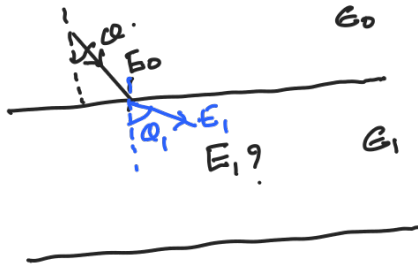
ie;

Boundary Condition ②
BC-2

Tangential components of electric field at dielectric boundary is continuous.



Example: Find $\vec{E}_1$ from the given data below,



1. Normal component of $\vec{D}$ is continuous.

$$\Rightarrow \epsilon_0 E_0 \cos \theta = \epsilon_1 E_1 \cos \theta_1 \quad \text{--- (1)}$$

2. tangential component of $\vec{E}$ is continuous.

$$\Rightarrow E_0 \sin \theta = E_1 \sin \theta_1 \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} \Rightarrow \epsilon_0^{-1} \tan \theta = \epsilon_1^{-1} \tan \theta_1 \quad \sim \text{Similar to Snell's law!}$$

(or in some sense it says the same thing)

θ_1 can be calculated.

$$\frac{(1)}{\epsilon_1} \Rightarrow \frac{\epsilon_0}{\epsilon_1} E_0 \cos \theta = E_1 \cos \theta_1 \quad \text{--- (3)}$$

$$(3)^2 + (2)^2 \Rightarrow \left(\frac{\epsilon_0}{\epsilon_1} \right)^2 E_0^2 \cos^2 \theta + E_0^2 \sin^2 \theta = E_1^2 \cos^2 \theta_1 + E_1^2 \sin^2 \theta_1$$

$$= E_1^2$$

* $E_0 \sin \theta$ (tangential component being continuous) contributes completely without any reduction to $\vec{E}_1$

* Normal component is reduced by $\frac{\epsilon_0}{\epsilon_1}$ times!

$$E_1 = E_0 \sqrt{\left(\frac{\epsilon_0}{\epsilon_1}\right)^2 \cos^2 \theta + \sin^2 \theta} \quad \text{--- (A)}$$

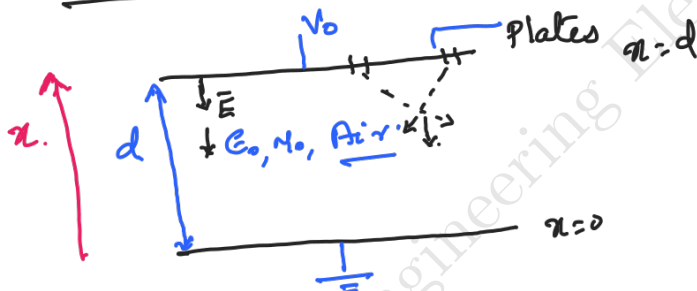
If $\theta = 0$ (normal incidence)
 $E_1 = E_0 \frac{\epsilon_0}{\epsilon_1}$

$$= E_0 \frac{\epsilon_0}{\epsilon_0 \epsilon_1} = \frac{E_0}{\epsilon_1} \rightarrow E_1 \text{ is reduced by the dielectric constant}$$

* From Eqn(A) we can see as ' θ ' increases, there can be a value of ' θ ' for which $E_0 = E_1 \Rightarrow$ matching condition.

Some more problems on capacitance.

Problem 1:

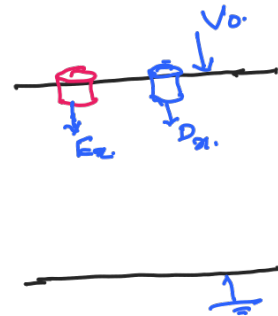


$$\vec{E} = E_x(x) \hat{x}$$

Now apply Gauss's law

$$E_x dA = \frac{\sigma dA}{\epsilon_0}$$

$$E_x = \frac{\sigma}{\epsilon_0}$$



Same idea we can try with displacement vectors

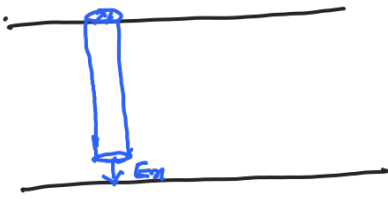
$$\nabla \cdot \mathbf{D} = \rho$$

with same Gaussian cylinder.

$$D_x dA = \sigma dA$$

$$D_x = \sigma$$





Irrespective of how large our gaussian surface is

$$E_n = \frac{\sigma}{\epsilon_0}$$

$D_n = \sigma$ i.e; E_n and D_n are constants.

Now voltage drop $V_0 = - \int_0^d \vec{E} \cdot d\vec{l} = \frac{\sigma}{\epsilon_0} d$.

Amount of charge on plate $\cdot Q = \sigma A$

i.e;

Voltage $\propto \sigma$

Total charge $\propto \sigma$

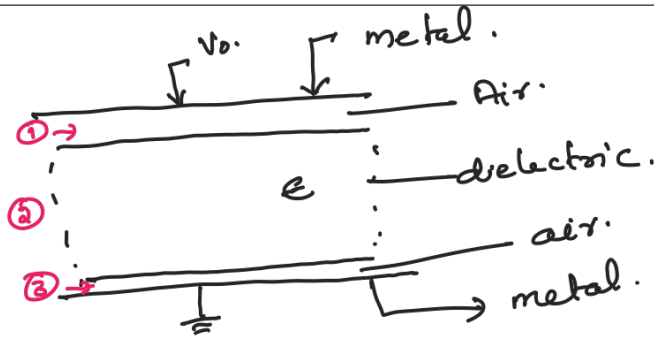
$$\frac{Q}{V_0} = \frac{\sigma A}{\sigma d / \epsilon_0} = \frac{\epsilon_0 A}{d} = C$$

$C \rightarrow$ depends only on device parameters $\rightarrow$ not on $\vec{E}$!
not on V !

Definition of capacitance: It is the measure of how much charge is induced on a plate when we put that plate at 1V w.r.t ground. Unit: $C/V = \text{Farad (F)}$

* Farad is very large number as 1 Coulomb is large number. (1C charge is very rare to see in any devices in real life)

Hence in capacitance usually we see range from pF to μ F.
(except for some special cases or specific applications).

Problem 2:

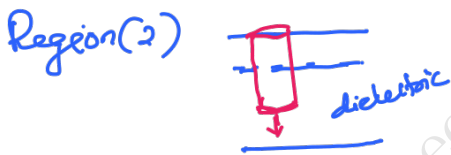
As we discussed before, we can use Gauss's law, however here we have induced charges on dielectric. So we use instead of $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$ $\nabla \cdot \mathbf{D} = \rho$.

$\oint \mathbf{D} \cdot d\mathbf{s}$: Charge enclosed



top wall metal
bottom wall inside air.
Side wall does not contribute.

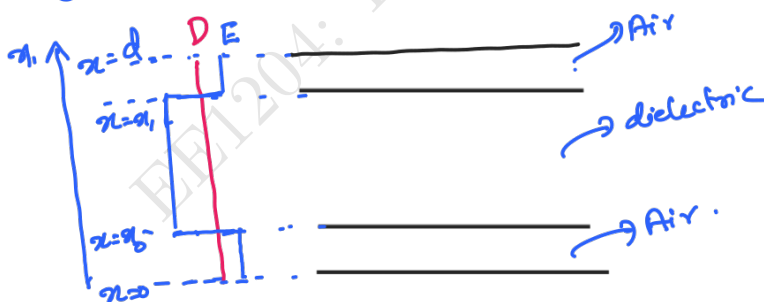
$$\Rightarrow D_1 dA = \sigma dA \quad D_{1x} = \sigma; \quad E_{1x} = \frac{\sigma}{\epsilon_0}.$$



→ top wall metal
bottom wall inside dielectric

$$D_2 dA = \sigma dA \quad D_{2x} = \sigma; \quad E_{2x} = \frac{\sigma}{\epsilon_0 \epsilon_r}$$

Region(3) $D_3 dA = \sigma dA \quad D_{3x} = \sigma; \quad E_{3x} = \frac{\sigma}{\epsilon_0}.$



Displacement vector
→ Uniform in 'x'

But electric field is
not uniform in 'x'.

* Here we know $\mathbf{D}$ & $\mathbf{E}$, but in terms of ϵ' (unknowns)!

Top plate is at V_0

$$V_0 = -\int_0^d \vec{E} \cdot d\vec{l} = \int_0^{x_0} \frac{\sigma}{\epsilon_0} dx + \int_{x_0}^{x_1} \frac{\sigma}{\epsilon} dx + \int_{x_1}^d \frac{\sigma}{\epsilon_0} dx.$$

$$V_0 = \frac{\sigma}{\epsilon_0} x_0 + \frac{\sigma}{\epsilon} (x_1 - x_0) + \frac{\sigma}{\epsilon_0} (d - x_1)$$

Charge $Q = \sigma A$

$$\frac{Q}{V_0} = \frac{A}{\frac{x_0}{\epsilon_0} + \frac{(x_1 - x_0)}{\epsilon} + \frac{(d - x_1)}{\epsilon_0}}$$

$$C = \frac{A}{\frac{x_0}{\epsilon_0} + \frac{(x_1 - x_0)}{\epsilon} + \frac{(d - x_1)}{\epsilon_0}}$$

} Again Capacitance depends only on material & the spacing with which we make the device!

(a) Now if we fill the entire gap with dielectric such that $x_0 = 0$; $x_1 = d$

$$\Rightarrow C = \frac{\epsilon A}{d}$$

(b) When dielectric shrinks to nothing.

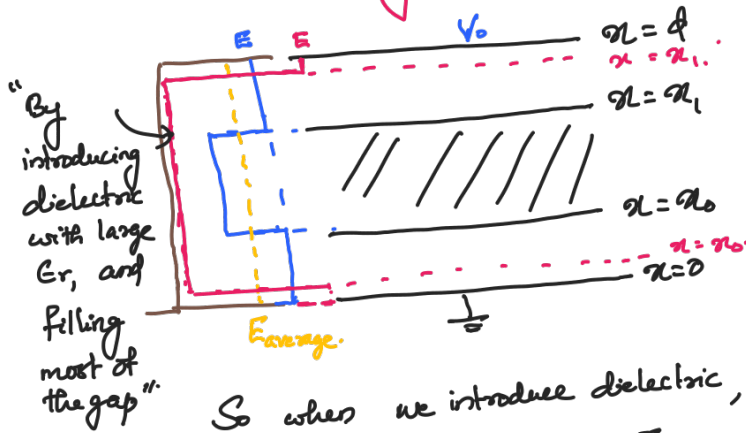
ie, $x_1 = x_0$

$$\Rightarrow C = \frac{\epsilon_0 A}{d}$$

ie, C increases from $\frac{\epsilon_0 A}{d} \rightarrow \frac{\epsilon_0 \epsilon_r A}{d}$ when we introduce dielectric.

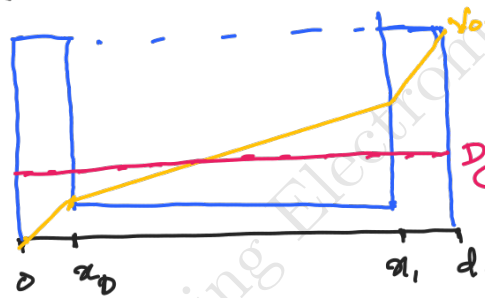
ie, As we increase the dielectric thickness here, the amount of stored charge for a given potential V_0 keeps increasing!

What is happening here?



We need an average $\bar{E}$, so that when we integrate $\int \bar{E} \cdot d\ell \rightarrow$ we build up V_0 .

So when we introduce dielectric, we get $\bar{E}$ field over those region less than $E_{average}$, and which is compensated by larger $\bar{E}$ field in the non-dielectric region near the plate $\Rightarrow$ a lot of induced free charge! $\Rightarrow$ increase capacitance.

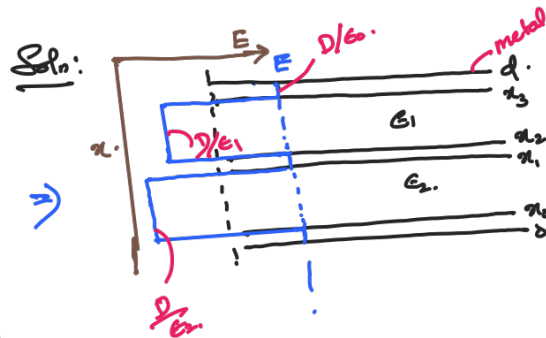
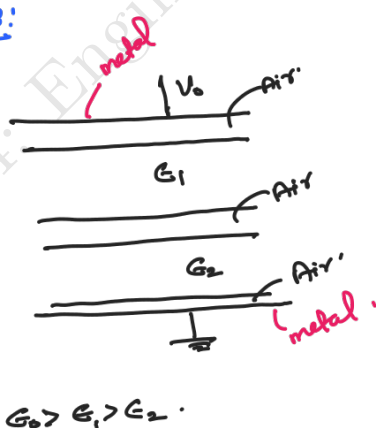


E — Electric field
 V — potential.
(Slope of the $\bar{V}$ is $\bar{E}$)

D (not in scale) is constant.

Again the concept of displacement vector helped us to understand this concept with ease.

Problem 3:



$$V_0 = \frac{\sigma}{\epsilon_0} \left[(x_0 - 0) + (x_2 - x_1) + (d - x_3) \right] + \frac{\sigma}{\epsilon_1} (x_3 - x_2) + \frac{\sigma}{\epsilon_2} (x_1 - x_0)$$

$$Q = \sigma A$$

$$C = \frac{Q}{V_0}$$



$$\text{Stored energy} = \frac{1}{2} \int \rho \phi \, dv = \frac{1}{2} \int \epsilon_0 E^2 \, dv$$

Still true.
 $\epsilon_0 \neq \epsilon$.

True only when
without dielectric
medium.

So when we have dielectric:

$$W = \frac{1}{2} \int \rho \phi \, dv \quad \left| \begin{array}{l} \nabla \cdot \bar{D} = \rho \\ \left[\frac{\partial(\rho \phi)}{\partial x} = \rho \frac{\partial \phi}{\partial x} + \phi \frac{\partial \rho}{\partial x} \right] \end{array} \right.$$

$$= \frac{1}{2} \int (\nabla \cdot \bar{D}) \phi \, dv.$$

$$\left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \phi \quad \rightarrow \quad \left(\frac{\partial \phi}{\partial x} \right) \rho = \frac{\partial(\rho \phi)}{\partial x} - \phi \frac{\partial \rho}{\partial x}$$

$$= \left(\frac{\partial}{\partial x} (D_x \phi) - D_x \frac{\partial \phi}{\partial x} \right) + \left(\frac{\partial}{\partial y} (D_y \phi) - D_y \frac{\partial \phi}{\partial y} \right) + \left(\frac{\partial}{\partial z} (D_z \phi) - D_z \frac{\partial \phi}{\partial z} \right)$$

$$= \nabla \cdot (\bar{D} \phi) - \bar{D} \cdot (-\bar{E})$$

$$= \nabla \cdot (\bar{D} \phi) + \bar{D} \cdot \bar{E}$$

$$\frac{1}{2} \int \rho \phi \, dv = \frac{1}{2} \int (\nabla \cdot \bar{D}) \phi \, dv$$

$$= \frac{1}{2} \int (\nabla \cdot (\bar{D} \phi) + \bar{D} \cdot \bar{E}) \, dv.$$

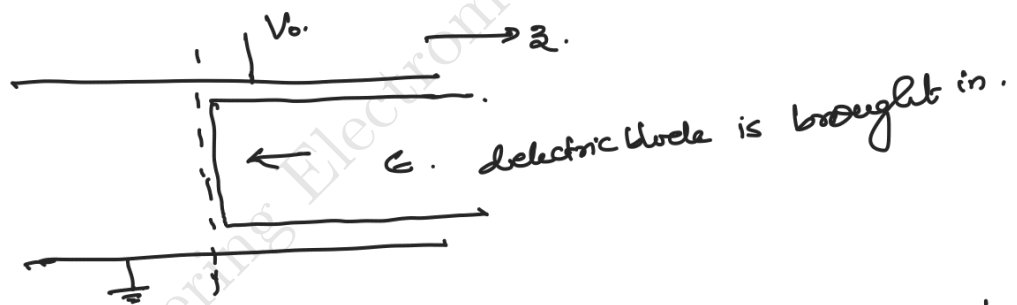
$$= \frac{1}{2} \oint \bar{D} \phi \cdot d\bar{s} + \boxed{\frac{1}{2} \int \bar{D} \cdot \bar{E} \, dv.} \rightarrow \text{energy in the field.}$$

This surface integral depends on what device we have.

$\frac{\vec{D} \cdot \vec{E}}{2} \Rightarrow$ is like energy density in the field.
 When dielectric is present we cannot use $\frac{1}{2} \epsilon_0 E^2$, rather
 we use $\frac{1}{2} \vec{D} \cdot \vec{E} = \epsilon \frac{\vec{E} \cdot \vec{E}}{2} \rightarrow$ stored energy increases
 by factor ϵ_r for fixed $(\vec{E})$
 $= \frac{\vec{D} \cdot \vec{D}}{2\epsilon} \rightarrow$ Decreases for fixed
 $\vec{D}$.

- * If $\vec{E}$ held constant stored energy $\uparrow$ with ϵ_r .
- * If $\vec{D}$ held constant stored energy $\downarrow$ with ϵ_r .

Example.



Given $V_0 \rightarrow \vec{E}$ constant $\Rightarrow$ stored energy increases by ϵ_r .

Instead of V_0 , if we charge the capacitor plate to Q
 and remove power supply $\Rightarrow$ Given Q constant
 $\Rightarrow \vec{D}$ constant
 $\Rightarrow$ stored energy decreases
 by ϵ_r if we introduce
 dielectric.